Objective
The objective of this practical is to perform advanced log analysis by correlating security events, detecting anomalies in network behavior, and enriching log data with contextual information to improve threat detection and investigation.

Tools Used
- Elastic Security (simulated)
- Security Onion (conceptual reference)
- Google Sheets / Excel (for implementation and visualization)

Log Correlation
Description
Log correlation was performed by linking failed login attempts with subsequent outbound network activity to identify potential attack patterns.

Correlated Log Data

Timestamp	Event ID	Source IP	Destination IP	Notes
2026-01-13 12:00:00	4625	192.168.1.100	8.8.8.8	Suspicious DNS request

Interpretation
The failed login event (Event ID 4625) followed by outbound traffic suggests a possible brute-force attempt or unauthorized access leading to suspicious network communication.

Anomaly Detection
Rule Definition
A rule was defined to detect abnormal outbound data transfer:
Condition:
If outbound traffic exceeds **1MB within 1 minute**, trigger an alert.

Simulated Detection Output

Timestamp	Source IP	Bytes Out	Alert Status
2026-01-13 12:01:00	192.168.1.100	1.4 MB	ALERT TRIGGERED

Interpretation
The system detected unusually high outbound traffic immediately after a failed login attempt. This behavior may indicate potential data exfiltration or compromised system activity, requiring further investigation.

Log Enrichment (GeoIP)
Description
GeoIP enrichment was applied to enhance log data by adding geographical context to IP addresses.

Enriched Log Data

IP Address	Country	City	ISP
8.8.8.8	USA	Mountain View	Google LLC

The destination IP address was identified as a legitimate Google DNS server located in the United States. This helps analysts understand whether the communication is expected or suspicious based on context.

GeoIP enrichment enhances log analysis by adding geographical and organizational context to IP addresses. In this case, the destination IP 8.8.8.8 was identified as a Google DNS server located in the United States. This improves visibility and helps analysts differentiate between legitimate and suspicious network activity.

SOC Relevance
This practical demonstrates key SOC capabilities:

- Correlating logs to identify attack patterns
- Detecting anomalies using rule-based monitoring
- Enriching logs with contextual intelligence
- Improving investigation accuracy and reducing false positives

Conclusion
Advanced log analysis was successfully performed by correlating failed login events with network activity, detecting anomalies in outbound traffic, and enriching logs with GeoIP data. This approach enables SOC analysts to identify potential threats, understand attack patterns, and respond effectively to security incidents.